Name of the Student: _____ **Roll no:** _____

CLASS: - T.E (CSE) **Division:** **Course: - PCC305CSE- Machine Learning Lab**

Experiment No. 05

**** Classify data using regularization techniques****

Marks: /10

Date of Performance: / /2026 **Sign with Date:**

Aim: To Classify data using regularization techniques such as Ridge and Lasso regression.

Lab Objective: To observe the influence on model overfitting and multicollinearity.

Problem Statement:

Classify data using regularization techniques such as Ridge and Lasso regression on a suitable multivariate open-access dataset and observe the influence on model overfitting and multicollinearity.

Theory:

1. Theoretical Background

overfitting and resolves **multicollinearity**.

1. Theoretical Background

When multiple input features are highly correlated (**multicollinearity**), the Ordinary Least Squares (OLS) covariance matrix (X^TX) becomes nearly singular, causing coefficient estimates (β) to fluctuate wildly with high variance. Regularization introduces a penalty term to constrain the weights.

1.1 Ordinary Least Squares (OLS) Cost Function

$$J_{OLS}(\beta) = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \|y - X\beta\|_2^2$$

1.2 Ridge Regression ($L2$ Regularization)

Ridge adds a penalty equal to the sum of the squared magnitude of coefficients:

$$J_{Ridge}(\beta) = \|y - X\beta\|^2_2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

Multicollinearity Handling: Shrinks correlated coefficients proportionally toward zero, stabilizing variance without dropping features entirely.

1.3 Lasso Regression (L_1 Regularization)

Lasso (Least Absolute Shrinkage and Selection Operator) adds a penalty proportional to the absolute values of coefficients:

$$J_{Lasso}(\beta) = \|y - X\beta\|^2_2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

Feature Selection: Due to the diamond geometry of the L_1 constraint, Lasso drives less informative or redundant collinear coefficients strictly to **zero**, performing automated feature selection.

2. Dataset Selection

- **Dataset: Diabetes Dataset** (Available in Scikit-Learn via load_diabetes or UCI ML Repository)
- **Samples (N):** 442 patients
- **Predictors (p):** 10 baseline physiological measurements: age, sex, bmi, bp, s1 (total cholesterol), s2 (LDL), s3 (HDL), s4 (total/HDL), s5 (serum triglycerides), s6 (blood glucose).
- **Multicollinearity Factor:** High natural collinearity among blood serum measurements (e.g., s1, s2, s3, s4, s5).
- **Target (Y):** Quantitative measure of disease progression one year after baseline.

3. Algorithm & Logic Flow

INPUT : Feature Matrix X, Target Vector y, Penalty parameter λ (alpha)
OUTPUT : OLS, Ridge, and Lasso parameter vectors (β), VIF diagnostic, Train/Test MSE & R^2

STEP 1: Load multivariate dataset and verify presence of collinearity.
STEP 2: Partition dataset into 80% Training and 20% Testing subsets.
STEP 3: Standardize features (StandardScaler) so penalties apply uniformly across units.
STEP 4: Fit Baseline OLS (LinearRegression) to observe high variance / coefficient inflation. STEP 5: Fit Ridge Regression across penalty range $\lambda \in [0.01, 100]$.
STEP 6: Fit Lasso Regression across penalty range $\lambda \in [0.01, 100]$.
STEP 7: Compare test performance (MSE, R^2) and count non-zero coefficients.

4. How Regularization Resolves Overfitting & Multicollinearity

1. **Multicollinearity Cushioning (Ridge):**

   ◦ Features s1 and s2 have a 0.90 correlation. In standard OLS, one receives a
   ◦ large positive weight while the other receives a large offsetting negative weight.

   Ridge penalizes large squares (β^2), forcing both to share reasonable, moderate weights and stabilizing the matrix calculation.

2. **Feature Sparsity & Redundancy Elimination (Lasso):**

   ◦ Lasso completely drops redundant correlated columns by driving 3 coefficients
   ◦ (s2, s4, age) to zero.

   This yields a sparser, more interpretable model that avoids overfitting while improving test performance (R^2 improves from 0.4526 to 0.4642).

Questions:

1. Why does Lasso set coefficients exactly to zero, while Ridge only shrinks them asymptotically?
2. What happens to Ridge Regression when λ=0 and when $\lambda \to \infty$?
3. Why is feature scaling mandatory before running Ridge or Lasso regression?
4. How does multicollinearity inflate variance in Ordinary Least Squares?**.** Answer: The closed-form OLS solution is β =
5. In what scenario would you choose Ridge over Lasso, and vice-versa?

Attendance and Timely Submission (10)	Execution (10)	Observations and Records (10)	Journal Documentation (10)	Oral (10)

Signature of Subject Teacher